Juliana Oyola-Pabon

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Summary

Software Engineer with a record of engineering enterprise Java backend services, high-scale data pipelines, and robust AWS cloud environments. Experienced in centralizing distributed media infrastructures hosting millions of active digital assets, architecting automated validation systems to protect downstream integrity, and optimizing data layers to yield measurable performance gains.

Skills & Tools

Languages: Java, Python, TypeScript, JavaScript, Swift, SQL, C++, HTML/CSS
Backend & Cloud: Spring Boot, Node.js, REST APIs, Microservices, Distributed Systems, ETL, AWS CDK, Lambda, API Gateway, S3, DynamoDB, RDS, CloudWatch, IAM
Data & Tools: PostgreSQL, MySQL, Docker, Terraform, Git, GitHub Actions, Jenkins, Linux, Maven, Gradle, JUnit, Mockito, React, Next.js

Professional Experience

Amazon Seattle, WA
Software Engineer | Prime Video *May 2024 – September 2025*

- Engineered core Java backend services and REST APIs to consolidate the metadata architecture for Prime Video, migrating millions of disparate content assets into a single unified publishing platform while decommissioning a network of fragmented legacy systems.
- Architected distributed ingestion pipelines and real-time validation workflows for incoming partner metadata, which normalized highly inconsistent structural schemas and isolated corrupted records before they could cause processing faults in downstream systems.
- Provisioned and managed repeatable production infrastructure deployments using AWS CDK, Lambda, API Gateway, DynamoDB, and S3, configuring explicit IAM policies and CloudWatch metrics to accelerate active incident response.
- Facilitated technical design reviews and code evaluations within the engineering group, authoring systemic architectural documentation and onboarding incoming technical staff to the backend deployment pipeline.

Amazon Seattle, WA
Software Engineer Intern | Prime Video *May 2023 – August 2023*

- Designed and deployed an automated validation and transformation workflow using Java and AWS CDK, which intercepted malformed partner data at the ingestion layer to eliminate roughly six hours of manual operational intervention per incident.
- Exercised end-to-end technical lifecycle ownership of this transformation microservice, executing its progression from initial system rules through to complete production monitoring instrumentation.

Amazon Seattle, WA
Software Development Engineer Intern | Amazon Care *May 2022 – August 2022*

- Programmed core Java backend features supporting critical scheduling capabilities, prescription logistics, and transactional message dispatch for a digital healthcare application handling more than 25,000 monthly active users.
- Restructured routing systems by substituting legacy chronological logic with proximity-based matching algorithms, optimizing logistical synchronization and contributing directly to a twofold rise in platform engagement alongside a 30% decrease in user drop-off.

Green Bank Observatory Green Bank, WV
Web Developer Intern | Skynet Robotic Telescope Network *January 2020 – May 2021*

- Managed the backend data layers supporting a robotic observation network, developing custom REST APIs to service thousands of astronomical processes, which compressed active storage requirements by 20% and accelerated query speeds by 30%.

Education

Florida State University Aug 2020 – May 2024
B.S. Computer Science

Santa Fe College August 2019 – August 2020
A.A. Psychology, Honors